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JEE Main Physics Practice Sheet — DPP-029

Topic: Static Friction — Concept, Laws of Friction & Angle of Friction

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A block of mass $m = 5$ kg rests on a rough horizontal surface with coefficient of static friction $\mu_s = 0.6$. A horizontal force of $F = 15$ N is applied to the block. The frictional force acting on the block is ($g = 10$ m/s^2):

(A) 30 N
(B) 15 N
(C) 0 N
(D) 50 N

Q2. The angle of static friction λ between two surfaces in contact is defined as the angle made by the resultant contact force with the:

(A) Direction of impending relative motion
(B) Normal reaction in the limiting friction state
(C) Horizontal surface
(D) Applied external pulling force

Q3. A block of mass $m = 2$ kg is pressed against a rough vertical wall by a horizontal force F. If the coefficient of static friction between the block and the wall is $\mu_s = 0.4$, the minimum force F required to prevent the block from slipping downward is ($g = 10$ m/s^2):

(A) 20 N
(B) 40 N
(C) 50 N
(D) 8 N

Q4. The coefficient of static friction between a block and a horizontal floor is μ_s. The minimum magnitude of force F applied at an angle θ to the horizontal required to just start moving the block of mass m is:

(A) $\dfrac{\mu_s mg}{\sqrt{1+\mu_s^2}}$
(B) $\dfrac{\mu_s mg}{\sqrt{1-\mu_s^2}}$
(C) $\mu_s mg$
(D) $\dfrac{mg}{\sqrt{1+\mu_s^2}}$

Q5. In Question 4, the optimal angle θ with the horizontal at which the pulling force F has its absolute minimum magnitude is equal to:

(A) $\tan^{-1}(\mu_s)$ (Angle of friction)
(B) $\sin^{-1}(\mu_s)$
(C) $\cos^{-1}(\mu_s)$
(D) $45°$

Q6. A block of mass M rests on a rough horizontal plane. When a horizontal force F is applied such that $F < f_s^{\max}$, the total contact force R exerted by the surface on the block is given by:

(A) Mg
(B) $\sqrt{(Mg)^2 + F^2}$
(C) $\sqrt{(Mg)^2 + (\mu_s Mg)^2}$
(D) $Mg + F$

Q7. A block of weight $W = 60$ N is placed on a horizontal table ($\mu_s = 0.5$). A horizontal force of 20 N is applied to it. The magnitude of the resultant reaction force exerted by the table on the block is:

(A) 60 N
(B) 20 N
(C) $20\sqrt{10}$ N
(D) $30\sqrt{5}$ N

Q8. Which of the following statements regarding the laws of static friction is INCORRECT?

(A) The limiting frictional force is directly proportional to the normal reaction
(B) The limiting static friction is independent of the apparent macroscopic area of contact
(C) Static friction is a self-adjusting force in both magnitude and direction
(D) The coefficient of static friction μ_s depends directly on the surface area in contact

Q9. A heavy block of mass 10 kg is resting on a rough horizontal floor with $\mu_s = 0.5$ and $\mu_k = 0.4$. A horizontal force of $F = 40$ N is applied. If the applied force is then increased to 60 N, the frictional force acting on the block in the two respective cases is ($g = 10$ m/s^2):

(A) 40 N and 40 N
(B) 40 N and 60 N
(C) 50 N and 40 N
(D) 50 N and 50 N

Q10. A uniform ladder of mass m rests against a smooth vertical wall with its foot on a rough horizontal floor (μ_s). The minimum angle θ that the ladder can make with the floor without slipping is:

(A) $\tan^{-1}\left(\dfrac{1}{2\mu_s}\right)$
(B) $\tan^{-1}\left(\dfrac{1}{\mu_s}\right)$
(C) $\tan^{-1}(2\mu_s)$
(D) $\tan^{-1}(\mu_s)$

Q11. A block of mass $m = 4$ kg is kept on a rough horizontal plane ($\mu_s = 0.75$). A force of $F = 20$ N acts downward at an angle of $37°$ to the horizontal pushing the block. The frictional force developed between the block and the floor is ($\sin 37° = 0.6, \cos 37° = 0.8, g = 10$ m/s^2):

(A) 16 N
(B) 39 N
(C) 28 N
(D) 20 N

Q12. If the angle of friction between two dry surfaces is $\lambda = 30°$, the maximum angle of an inclined plane made with the horizontal on which a body can rest without sliding down (angle of repose ϕ) is:

(A) $60°$
(B) $30°$
(C) $45°$
(D) $15°$

Q13. A block of mass $m = 1$ kg is placed against the front vertical wall of a cart. The coefficient of static friction between the block and the cart wall is $\mu_s = 0.2$. The minimum horizontal acceleration a of the cart required to prevent the block from falling is ($g = 10$ m/s^2):

(A) 2 m/s^2
(B) 20 m/s^2
(C) 50 m/s^2
(D) 5 m/s^2

Q14. A block of mass $m = 3$ kg is held in equilibrium on a rough horizontal plane by two perpendicular horizontal forces $F_1 = 6$ N and $F_2 = 8$ N. If $\mu_s = 0.5$, the magnitude of the static friction force acting on the block is ($g = 10$ m/s^2):

(A) 15 N
(B) 10 N
(C) 14 N
(D) 2 N

Q15. Two blocks of masses $m_1 = 2$ kg and $m_2 = 4$ kg are connected by a light string on a rough horizontal table with $\mu_s = 0.3$ for both blocks. A horizontal pulling force $F = 12$ N is applied to m_2. The tension in the connecting string is ($g = 10$ m/s^2):

(A) 4 N
(B) 6 N
(C) 0 N
(D) 12 N

Q16. A box of mass M is resting on the flatbed of a truck. The coefficient of static friction between the box and the bed is $\mu_s = 0.4$. The maximum acceleration of the truck so that the box does not slip backwards relative to the truck is ($g = 10$ m/s^2):

(A) 4 m/s^2
(B) 2.5 m/s^2
(C) 10 m/s^2
(D) 0.4 m/s^2

Q17. An insect crawls up the inside of a rough hemispherical bowl of radius R. If the coefficient of static friction between the insect and the bowl is $\mu_s = 0.75$, the maximum height h to which the insect can crawl above the lowest point without slipping is:

(A) $0.2R$
(B) $0.4R$
(C) $0.6R$
(D) $0.8R$

Q18. A body of mass m is placed on a rough horizontal plane (μ_s). A pulling force P making an angle θ above the horizontal acts on the body. If the body is just on the verge of sliding, the normal reaction from the ground is:

(A) $mg - P\sin\theta$
(B) $mg + P\sin\theta$
(C) $mg\cos\theta$
(D) $\dfrac{mg}{\cos\theta}$

Q19. When a person walks forward on a horizontal ground, the direction of the static frictional force exerted by the ground on the person's foot is:

(A) In the backward direction
(B) In the forward direction
(C) Vertically upward
(D) Zero

Q20. A block of mass m is placed on an inclined plane whose inclination θ is continuously increased from $0°$. The graph showing the variation of frictional force f acting on the block with angle of inclination θ until and past the angle of repose $\phi = \tan^{-1}\mu_s$ is:

(A) Increases as $mg\sin\theta$ for $\theta \le \phi$, then suddenly drops to $\mu_k mg\cos\theta$ and decreases
(B) Constant at $\mu_s mg$ for all $\theta \le \phi$
(C) Decreases as $mg\cos\theta$, then increases
(D) Linearly increases throughout with slope μ_s

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-029)

Static Friction, Laws of Friction & Angle of Friction

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	A	16	A
2	B	7	C	12	B	17	A
3	C	8	D	13	C	18	A
4	A	9	A	14	B	19	B
5	A	10	A	15	C	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (B)
Normal reaction $N = mg = 5 \times 10 = 50$ N.
Maximum limiting static friction:

$$f_s^{\max} = \mu_s N = 0.6 \times 50 = 30 \text{ N}$$

Since the applied horizontal force $F = 15 \text{ N} < f_s^{\max}$, the block does not move.
Static friction is self-adjusting up to its limiting value:

$$f_s = F_{\text{applied}} = 15 \text{ N}$$

Sol 2. Option (B)
The angle of friction λ is defined as the angle between the resultant contact force $\vec{R} = \vec{N} + \vec{f}_s^{\max}$ and the normal reaction $\vec{N}$ in the limiting equilibrium state:

$$\tan\lambda = \frac{f_s^{\max}}{N} = \frac{\mu_s N}{N} = \mu_s$$

Sol 3. Option (C)
Horizontal equilibrium on the vertical wall gives: $N = F$.
For the block not to slip down, the upward static friction must balance weight:

$$f_s = mg \le f_s^{\max} = \mu_s N$$

$$mg \le \mu_s F \implies F \ge \frac{mg}{\mu_s} = \frac{2 \times 10}{0.4} = 50 \text{ N}$$

Hence, $F_{\min} = 50$ N.

Sol 4. Option (A)
When force F is applied at angle θ above the horizontal: Vertical equilibrium: $N + F\sin\theta = mg \implies N = mg - F\sin\theta$.
To initiate motion along the horizontal:

$$F\cos\theta = f_s^{\max} = \mu_s N = \mu_s(mg - F\sin\theta)$$

$$F(\cos\theta + \mu_s\sin\theta) = \mu_s mg \implies F = \frac{\mu_s mg}{\cos\theta + \mu_s\sin\theta}$$

The maximum value of the denominator is $\sqrt{1+\mu_s^2}$.
Thus, the minimum required force is:

$$F_{\min} = \frac{\mu_s mg}{\sqrt{1+\mu_s^2}}$$

Sol 5. Option (A)
From Solution 4, the denominator $D(\theta) = \cos\theta + \mu_s\sin\theta$ is maximized when:

$$\frac{dD}{d\theta} = -\sin\theta + \mu_s\cos\theta = 0 \implies \tan\theta = \mu_s$$

This optimal angle $\theta = \tan^{-1}\mu_s = \lambda$ (angle of friction).

Sol 6. Option (B)
The contact force $\vec{R}$ is the vector resultant of the normal reaction $\vec{N}$ and static friction $\vec{f}_s$.
Since $\vec{N}$ is vertical ($N = Mg$) and $\vec{f}_s$ is horizontal ($f_s = F$), they are mutually perpendicular:

$$R = \sqrt{N^2 + f_s^2} = \sqrt{(Mg)^2 + F^2}$$

Sol 7. Option (C)
Normal reaction $N = W = 60$ N.
Limiting friction $f_s^{\max} = \mu_s N = 0.5 \times 60 = 30$ N.
Since $F = 20 \text{ N} < 30$ N, the block is stationary and $f_s = 20$ N.
Resultant contact force:

$$R = \sqrt{N^2 + f_s^2} = \sqrt{60^2 + 20^2} = \sqrt{3600 + 400} = \sqrt{4000} = 20\sqrt{10} \text{ N}$$

Sol 8. Option (D)
According to the empirical laws of friction (Amontons' laws), the limiting static friction and coefficient μ_s depend only on the nature of the materials in contact and surface roughness, but are independent of the apparent macroscopic contact area. Hence statement (D) is incorrect.

Sol 9. Option (A)
Normal reaction $N = mg = 100$ N.
Limiting friction $f_s^{\max} = \mu_s N = 0.5 \times 100 = 50$ N.
Kinetic friction $f_k = \mu_k N = 0.4 \times 100 = 40$ N.
Case 1: $F = 40 \text{ N} < 50 \text{ N} \implies$ block remains at rest, $f = F = 40$ N.
Case 2: $F = 60 \text{ N} > 50 \text{ N} \implies$ block starts accelerating, dynamic state with kinetic friction $f = f_k = 40$ N.

Sol 10. Option (A)
Let ladder have length L and mass m.
Floor exerts $N_1 = mg$ and friction $f_s \le \mu_s N_1 = \mu_s mg$.
Smooth wall exerts horizontal normal reaction $N_2 = f_s$.
Taking torque about the bottom foot of the ladder at equilibrium:

$$N_2(L\sin\theta) - mg\left(\frac{L}{2}\cos\theta\right) = 0$$

$$N_2\sin\theta = \frac{mg}{2}\cos\theta \implies f_s = \frac{mg}{2\tan\theta}$$

For no slipping, $f_s \le \mu_s mg \implies \frac{mg}{2\tan\theta} \le \mu_s mg \implies \tan\theta \ge \frac{1}{2\mu_s}$.
Thus, $\theta_{\min} = \tan^{-1}\left(\frac{1}{2\mu_s}\right)$.

Sol 11. Option (A)
For the pushed block: Vertical normal force: $N = mg +$

$F \sin 37° = 4(10) + 20(0.6) = 40 + 12 = 52$ N.
Limiting static friction: $f_s^{\max} = \mu_s N = 0.75 \times 52 = 39$ N.
Applied horizontal pushing component: $F_x = F \cos 37° = 20(0.8) = 16$ N.
Since $F_x = 16 \text{ N} < 39$ N, the block does not slide and $f_s = F_x = 16$ N.

Sol 12. Option (B)
The angle of repose ϕ is mathematically equal to the angle of friction λ:

$$\tan \phi = \mu_s = \tan \lambda \implies \phi = \lambda = 30°$$

Sol 13. Option (C)
Normal reaction between block and front cart wall is the pseudo-force/contact force: $N = ma$.
Vertical static friction prevents downward fall: $f_s = mg \leq \mu_s N = \mu_s ma$.

$$mg \leq \mu_s ma \implies a \geq \frac{g}{\mu_s} = \frac{10}{0.2} = 50 \text{ m/s}^2$$

Sol 14. Option (B)
Resultant applied horizontal force:

$$F_{\text{net}} = \sqrt{F_1^2 + F_2^2} = \sqrt{6^2 + 8^2} = 10 \text{ N}$$

Limiting friction: $f_s^{\max} = \mu_s mg = 0.5 \times 3 \times 10 = 15$ N.
Since $F_{\text{net}} = 10 \text{ N} < 15$ N, static friction adjusts to exactly equal 10 N directed opposite to the resultant applied force.

Sol 15. Option (C)
Limiting friction on m_2 alone: $f_{s2}^{\max} = \mu_s m_2 g = 0.3 \times 4 \times 10 = 12$ N.
The applied force on m_2 is $F = 12$ N, which is completely balanced by the static friction beneath m_2 itself ($f_2 = 12$ N).
Hence, no tendency of relative displacement is transferred to the string, so Tension $T = 0$ N.

Sol 16. Option (A)
The only horizontal force providing forward acceleration to the box without slipping is static friction:

$$f_s = Ma \leq f_s^{\max} = \mu_s Mg$$

$$a_{\max} = \mu_s g = 0.4 \times 10 = 4 \text{ m/s}^2$$

Sol 17. Option (A)
At an angle α from the vertical center line, the normal reaction is $N = mg \cos \alpha$ and tangential gravity component is $mg \sin \alpha$.
For no slipping: $\tan \alpha \leq \mu_s = 0.75 = 3/4 \implies \cos \alpha = \frac{4}{5} = 0.8$.
Height above lowest point:

$$h = R(1 - \cos \alpha) = R(1 - 0.8) = 0.2R$$

Sol 18. Option (A)
Vertical equilibrium of the body gives:

$$N + P \sin \theta = mg \implies N = mg - P \sin \theta$$

Sol 19. Option (B)
While walking forward, the foot pushes the ground backwards. To oppose the backward slipping tendency of the foot, static friction from the ground acts in the forward direction on the foot, propelling the person forward.

Sol 20. Option (A)
For $\theta \leq \phi$, the block remains stationary, and self-adjusting static friction balances component of gravity: $f_s = mg \sin \theta$ (increasing sinusoidally). At $\theta = \phi$, it reaches $f_s^{\max}$. For $\theta > \phi$, the block slides and kinetic friction acts: $f_k = \mu_k mg \cos \theta$ (which decreases monotonically as θ increases).